

Education

University of Utah
B.S. Computer Science

Salt Lake City, UT
August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies
Associate Software Engineer

Dallas, TX
June 2025 – Present

- Owned signal processing modules end to end on embedded defense systems, from problem definition through deployment, working directly with hardware and firmware engineers to translate ambiguous constraints into working software with no formal spec.
- Built microservices and automation tooling from scratch for high-frequency sensor data ingestion, identifying and closing gaps in CI/CD and testing coverage on my own initiative.
- Delivered production releases under strict latency and resource constraints, communicating tradeoffs directly to hardware stakeholders throughout each ambiguous, fast-moving project.

Doxy.me
Software Engineer

Charleston, SC
August 2024 – May 2025

- Owned a real-time mental health inference system end to end, from problem definition through production, fine-tuning Llama models and deploying via SageMaker and Fargate over live WebRTC clinical sessions with no existing template to build from.
- Restructured the inference pipeline mid-build to absorb a new clinical requirement without a rewrite, keeping delivery on schedule.
- Closed the loop between model output and clinician action by building the TypeScript frontend surfacing AI-generated insights at the point of care, owning the full stack from model to UI.

University of Utah
Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT
May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

HEXstream
Software Engineering Intern

Chicago, IL
May 2022 – Aug 2022

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
 - Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.
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Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.